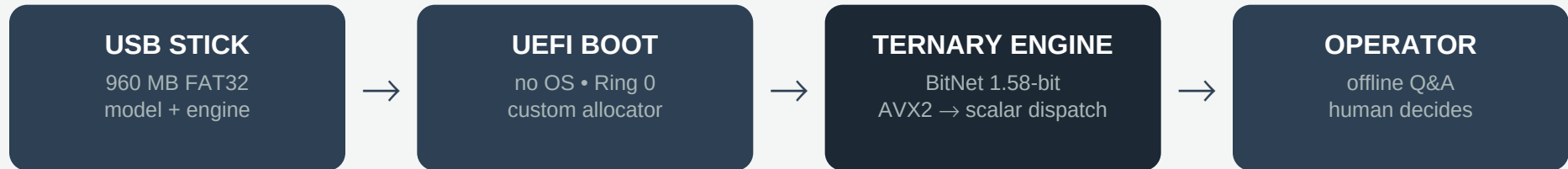


A.L.I.C.E. — Sovereign Bare-Metal AI for DDIL Operations

DARPA-SN-26-97 Low Resource Computing | 1(b) Low Memory + 2(d) Low Complexity of UX

ARCHITECTURE — THE OPERATING SYSTEM IS DELETED, NOT OPTIMIZED



MEASURED ON REAL LEGACY HARDWARE — EVERY NUMBER RUNTIME-COMPUTED AND LOGGED

< 1.5 GB

TOTAL PHYSICAL FOOTPRINT

782 MB read-only weights
+ 545.41 MB peak working memory
(runtime high-water mark)
OS footprint: 0 bytes

0.61 tok/s

ON A 2015 SCAVENGED LAPTOP

Broadwell i5, bare metal, single thread,
while firmware held the clock at 22%
of nominal — 5.08× SIMD dispatch
measured on the same boot

0 bytes

OS • NETWORK STACK • PERSISTENCE

no attack surface to patch,
no telemetry, no emissions,
host disk never mounted —
nothing left behind

ROADMAP: model portfolio to sub-1 GB (Falcon-E port in progress) → distilled DDIL mission model (published BitDistill recipe) → embedded x86 SoC targets

Compatibility floor (assessed): any 64-bit UEFI machine (~2012+) with 2 GB RAM — CPU generation sets speed, never boot success. Same binary: SSE2 scalar → AVX2+FMA.

Provenance: all figures computed at runtime by the engine's /gauntlet instrumentation, logged to the boot volume, harvested to a versioned dataset (git commit 6f29c6b).

Model: Microsoft BitNet b1.58-2B-4T (open, MIT). Systems contribution: the no_std bare-metal inference stack. Engine performs no targeting, no fire control, no transmission.